Monthly Intelligence Report

Edition 002 — The Marmara-İstanbul Corridor

Where the energy transition is outrunning the grid. edition · May 2026 · SSI v4.0.2

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SECTION A

Executive Summary

SUBSTATIONS SCORED

4,092

1,132 HV · 2,960 MV

GRID LINES MAPPED

14,221

~46,000 km coverage

MC SIMULATIONS

40.92 M

10,000 per substation

PUBLIC DATA SOURCES

30

95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). Its engine processes 146,000+ source data points per run, executes 42.9 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 1.78 billion arithmetic operations — producing 244,701 output data points with full uncertainty quantification across 4,092 substations and 14,221 grid lines spanning ~46,000 km.

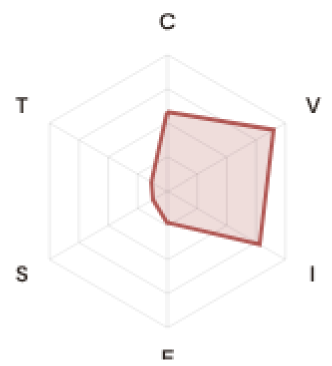
Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 30 verified public sources, making the methodology fully auditable and reproducible. Initially covering Turkey's entire transmission and distribution grid, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

Mersin-I TM

Mersin, Mediterranean · 66000 kV

0.583	High	0.161	26th
R_FINAL	CLASSIFICATION	CI WIDTH	FLEET PERCENTILE



COMPONENTS

- C Continuity: 0.580 / 0.30 (193%)
- I Infrastructure: 0.781 / 0.25 (313%)
- S Saturation: 0.120 / 0.20 (60%)
- V Voltage: 0.900 / 0.10 (900%) ⚠
- E Economic: 0.230 / 0.10 (230%)
- T Transition: 0.133 / 0.05 (266%)

MODIFIERS

- **R3 Consequence:** 1.061 ↑ amplifies
- **R4 Graph Criticality:** 1.010 → neutral
- **R6a Restoration:** 1.012 ↑ amplifies
- **R7 Digital Readiness:** 1.004 → neutral

This month's spotlight — automatically selected for May 2026 — is **Mersin-I TM** in Mersin, Mediterranean. With an R_final of 0.583 (High band, 26th fleet percentile), its risk profile is dominated by the V (Voltage) component at 900% of its weight ceiling, followed by I (Infrastructure) at 313%. Modifiers collectively shift R_base by 11.8%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that Turkey's grid modernisation investment planning requires.

CYBER HIGH EXPOSURE

2491

60.9% of fleet

BLIND SPOTS

1313

High/Critical + R7 < median

AVG R7 MODIFIER

1.0058

Range [0.99, 1.05]

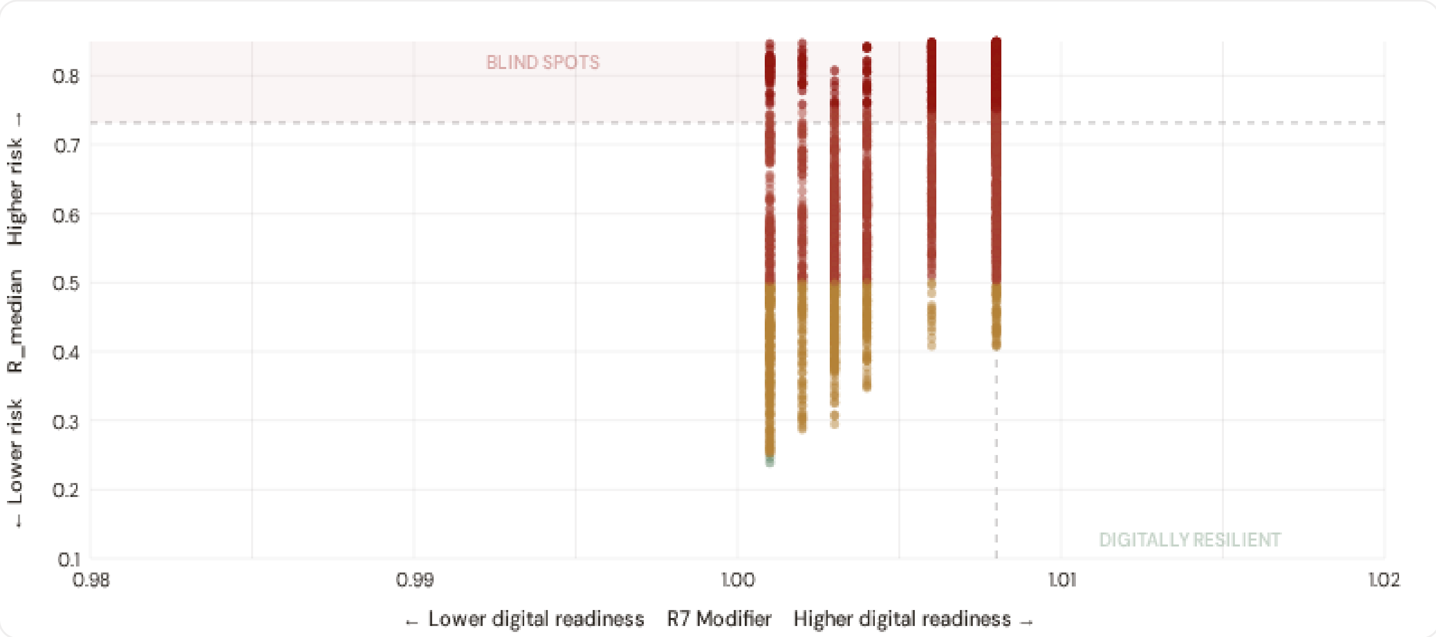
HIGH-CONSEQUENCE SUBS

2,214

54.1% · R3 ≥ 1.12

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



● Low ● Medium ● High ● Critical

Dashed lines = fleet medians

The cyber-physical matrix identifies **1313 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness (R7 < 1.0080). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Aegean (389), Central Anatolia (344), Mediterranean (276)**. Across the full fleet, 2491 substations (60.9%) carry a HIGH cyber-exposure classification, while 61 (1.5%) are in the LOW tier — indicating strong digital infrastructure coverage but concentrated in the northern urban grid.

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.

🏭 Capital-Intensive / High-Consequence

Est. VoLL: ₺15–30/kWh

Heavy industry, petrochemical, large manufacturing — highest economic loss per interruption

2,214 substations (54.1%) High/Critical: **2162** (97.7%) Avg R: **0.817** Avg E: **0.560** / 0.10

Low 0 Med 52 High 578 Crit 1584

🔧 Industrial / Medium-Large Enterprise

Est. VoLL: ₺8–15/kWh

Diversified industry, logistics, food processing — significant continuity requirements

559 substations (13.7%) High/Critical: **515** (92.1%) Avg R: **0.712** Avg E: **0.386** / 0.10

Low 0 Med 44 High 277 Crit 238

🏪 Commercial / SME-Dense

Est. VoLL: ₺3–8/kWh

Service economy, retail, tourism — moderate individual impact but high aggregate exposure

711 substations (17.4%) High/Critical: **494** (69.5%) Avg R: **0.564** Avg E: **0.218** / 0.10

Low 0 Med 217 High 456 Crit 38

🌾 Light / Agricultural / Rural

Est. VoLL: ₺1–3/kWh

Sparse economic activity, agricultural land — lower VoLL but higher social vulnerability

608 substations (14.9%) High/Critical: **304** (50.0%) Avg R: **0.529** Avg E: **0.063** / 0.10

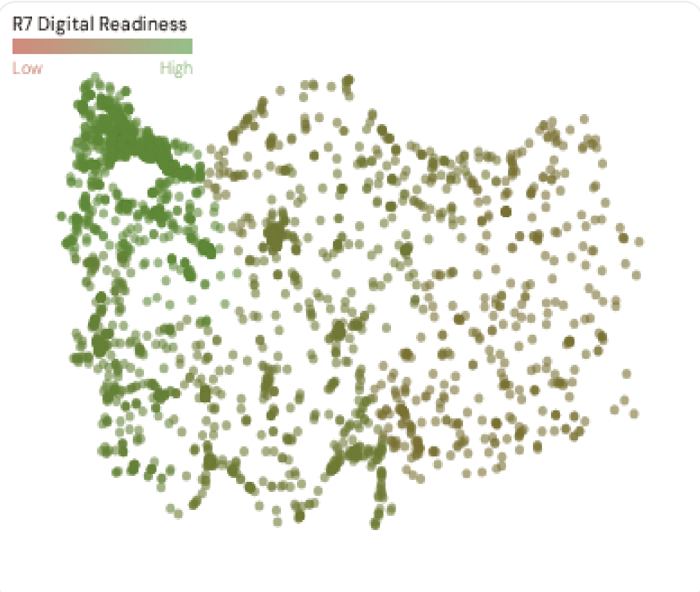
Low 2 Med 302 High 234 Crit 70

Value of Lost Load (VoLL) context: ACER's 2023 estimate for Turkey places average VoLL at ₺11.4/kWh for industrial customers and ₺3.2/kWh for residential. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **2162 substations** combine capital-intensive economic fabric (R3 ≥ 1.12) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **Marmara (2214)** — regions where industrial corridors depend on uninterrupted power for continuous processes (steel, glass, automotive). A single hour of unplanned outage at these substations carries an estimated VoLL of ₺15–30/kWh, compared to ₺1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 10.9%, but this masks sharp regional variation: Eastern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 Province Digital Readiness Ranking

Provinces ranked by average R7 modifier — lower values indicate weaker digital infrastructure. Cross-referenced with average R_median to identify provinces combining digital exposure with high grid stress.



WORST DIGITAL READINESS — TOP 15 PROVINCES

Longer bar = higher R7 = better digital infrastructure.

1	Kahramanmaraş	1.0010
2	Gümüşhane	1.0010
3	Bitlis	1.0010
4	Adıyaman	1.0010
5	Tunceli	1.0010
6	Mardin	1.0010
7	Ardahan	1.0010
8	Batman	1.0010
9	Şırnak	1.0010
10	Siirt	1.0010
11	Van	1.0010

12	Iğdır		1.0010
13	Bingöl		1.0010
14	Elazığ		1.0010
15	Rize		1.0010

Province-level analysis reveals a clear West-East digital divide in broadband and smart metering coverage. **Kahramanmaraş** has the lowest average R7 (1.0010), while **Tekirdağ** leads at 1.0080 — a spread of 0.0070 across the R7 range. **10 provinces** combine below-median digital readiness with above-median grid risk, making them priority targets for grid digitalisation investment. The R7 modifier is intentionally narrow ([0.99, 1.05]) to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

> **Inaugural edition — Baseline Snapshot (April 2026).** This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 1.0058, median = 1.0080; 2491 substations classified HIGH cyber-exposure; 1313 blind-spot substations (High/Critical risk + below-median R7); 346 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline, flagging provinces where digital infrastructure improvements (broadband rollout, smart meter deployment, OT upgrades) translate into measurable R7 shifts. The next material R7 update is expected when Eurostat publishes the 2025 DESI sub-indices (anticipated Q3 2026).

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **30 verified public data sources** feeding 95 variables across 6 components and 5 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES
30
95 variables

WEEKLY / MONTHLY
3
High-frequency feeds

QUARTERLY / ANNUAL
17
Regular refresh cycle

STATIC / DERIVED
3
Standards + scenarios

Data Source Registry — May 2026

TEIASPLAN	TEİAŞ Network Development Plans	MONTHLY	TSO zone	2 vars
AFAD-GEO	AFAD Geospatial Hazard Atlas	CONTINUOUS	Province	1 var
EDAS-OSM	EDAŞ Companies + OSM Power Infrastructure	QUARTERLY	Node/edge	3 vars
AEDAS-DGSI	DSO Alliance (AEDAS) + General Directorate Subtrans	QUARTERLY	Province	2 vars
EGOV-REG	E-Government Vehicle Registry (TUIK)	QUARTERLY	Province	1 var
TUBITAK-R	TÜBİTAK Innovation Enrichment (HRST)	QUARTERLY	Province	1 var
ETKB-REN	ETKB Renewable Energy Action Plans (YEKP)	QUARTERLY	Province	2 vars
EPDK	EPDK (Enerji Piyasası Düzenleme Kurumu)	ANNUAL	Province	8 vars
TUIK	TÜİK (Türkiye İstatistik Kurumu)	ANNUAL	Province	8 vars
TUBITAK	TÜBİTAK Innovation Registry	ANNUAL	Province	4 vars
ETKB	ETKB (Enerji ve Tabii Kaynaklar Bakanlığı)	ANNUAL	Province	5 vars
EPDK-MON	EPDK Monitoring & Enforcement Reports	ANNUAL	DSO-level	2 vars
ENRJ-ENV	Environmental & Air Quality Portal	ANNUAL	~5 km + station	3 vars
ILOSTAT	ILO Statistics (Energy Sector Employment)	ANNUAL	National	2 vars
DESI-TR	Turkey Digital Index (Regional DESI analogue)	ANNUAL	Regional	2 vars
HAZINE	Hazine ve Maliye Bakanlığı (Fiscal Data)	ANNUAL	Province	1 var
NUFUS	TURKSTAT Population Registry	ANNUAL	Province	2 vars
METROPOL	Metropolitan Municipalities Data Portal	ANNUAL	Province	2 vars
TUBITAK-ASTI	TÜBİTAK ASTI Energy Transition Roadmap	ANNUAL	Province	1 var
CBBB	Central Bank Blue Book (Economic Data)	ANNUAL	Regional	1 var
AFAD	AFAD (Afet ve Acil Durum Yönetimi Başkanlığı)	STATIC	Province	3 vars
CEDA	CEDA Energy Data Archive (Copernicus)	STATIC	0.25° (~25 km)	4 vars
IEEE-1	IEEE C57.91 / IEC 60076 / TBDY 2018	STATIC	Asset-level	16 vars
GSHHGM	General Directorate of Seismic Affairs	STATIC	~5 km	2 vars
COGIL	Central Address Registry (Merkezi Adresleme)	STATIC	Province	1 var
ISO-9223	ISO 9223 Corrosion (Turkey coastal analogue)	DERIVED	Province	1 var

TEIASDAT	TEİAŞ Transparency Portal (Veri Analiz Sistemi)	HOURLY	Bidding zone	3 vars
EPIAS	EPİAŞ / EXIST (Enerji Piyasaları İşletme A.Ş.)	HOURLY	Bidding zone	2 vars
MGM	MGM (Meteoroloji Genel Müdürlüğü)	HOURLY	~1 km	4 vars
GBML	GBML (Gözlemci Ağı – Ground-Based Monitoring)	HOURLY	~25 km	1 var
OMT	Open-Meteo Historical Weather	HOURLY	~1 km	3 vars

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS

Monthly (3 sources) — Grid load + procurement data

ETKB · TEIASPLAN · EPDK-MON

Quarterly (5 sources) — DER registry + business data + macro indicators

EDAS-OSM · EGOV-REG · AEDAS-DGSI · TUBITAK-R · ETKB-REN

Annual (12 sources) — Core grid quality + demographics + environment

EPDK · TUIK · TUBITAK · ETKB · HAZINE · NUFUS · METROPOL · DESI-TR · TUBITAK-ASTI · CBBB · ILOSTAT · ENRJ-ENV

Static (5 sources) — Standards + scenarios + seismic hazard

IEEE-1 · CEDA · GSHHGM · AFAD · COGIL

C.2 Fleet Score Snapshot

FLEET SIZE
4,092
substations scored

MEDIAN R
0.732
P5=0.405 · P95=0.962

HIGH + CRITICAL
3475
84.9% of fleet

HIGH CONFIDENCE
100%
4092 subs · CI/R < 0.50

BAND DISTRIBUTION

Low 2 (0.0%) · Medium 615 (15.0%) · High 1545 (37.8%) · Critical 1930 (47.2%)

No primary data sources have been updated since the v4.0.2 launch baseline. All **4,092 substation scores remain unchanged** this edition. The fleet median R of 0.732 (mean 0.716) reflects a distribution skewed toward the lower bands, with 0.0% in Low and 15.0% in Medium. The first material score changes are expected when EPDK publishes the annual electricity quality tables (affecting C1–C4 and R6a) and when EPDK refreshes the licensed generation registry (affecting T1). High-frequency sources (OSM, Open-Meteo) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2

17 changes tracked across P0, P1, and P2 releases

F1	NEW	New T component — Energy Transition Exposure (T1)	§2, §4
F2	NEW	New R6a — Restoration Speed Modifier (CAIDI-based)	§5
F3	ENHANCED	R3 enhanced — Energy Poverty Vulnerability (V_socio)	§5
F4	ENHANCED	R4 enhanced — Graph-theoretic betweenness + bridge detection	§5
F5	ENHANCED	R2 enhanced — Climate Trajectory (CMIP6 SSP2-4.5)	§5
F6	ENHANCED	R7 Digital Readiness — Province-level DESI-TR model	§5
L1	NEW	New I5, I7–I9 metrics — thermal, load, corrosion, seismic	§2, §8
L2	ENHANCED	E2 Innovation Enrichment — HRST + startup density (TÜBİTAK)	§6
L3	ENHANCED	V_socio Fiscal Enrichment — Hazine + METROPOL + energy price	§5
L4	ENHANCED	R3 Demographic Shift Amplifier — TÜİK migration patterns	§5
L5	DATA	95/95 variables operational (100%). 30+ data sources total.	§8, §12
G1	DATA	ETKB Registry upgraded to quarterly — Province-level DER registry	§12
G2	DATA	AFAD upgraded to live API — Province-level seismic/geological data	§12
G3	DATA	OSM upgraded to Overpass API — 4,092 real substations (Turkey)	§12
G4	NEW	R6b Network Topology modifier — centrality + ring analysis from OSM graph	§5
G5	NEW	Network & Topology layer (H): 7 variables — TEİAŞ Plans, ring analysis	§12
G6	NEW	Output Scores layer (I): 7 variables — risk_score, ETTC, stationary probs	§12

SECTION D — MONTHLY DEEP-DIVE

The Marmara-İstanbul Corridor: Where the Energy Transition Is Outrunning the Grid

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** Marmara corridor · **002** DER saturation hotspots · **003** Climate vulnerability under CMIP6 · **004** North-South economic impact disparity · **005** Voltage quality and industrial output · **006** Infrastructure age vs investment · **007** Stress–Resilience quadrant trends · **008** T-component forward projections · **009** Provincial Grid Health Cards · **010** Academic validation and peer feedback · **011** European replication feasibility · **012** Year in review.

D.1 Why Marmara, Why Now

MARMARA SUBSTATIONS	HIGH BAND	CORRIDOR MEDIAN R

2214	578 + 1584	0.839
396 HV · 1818 MV	97.7% of corridor	Fleet median: 0.732

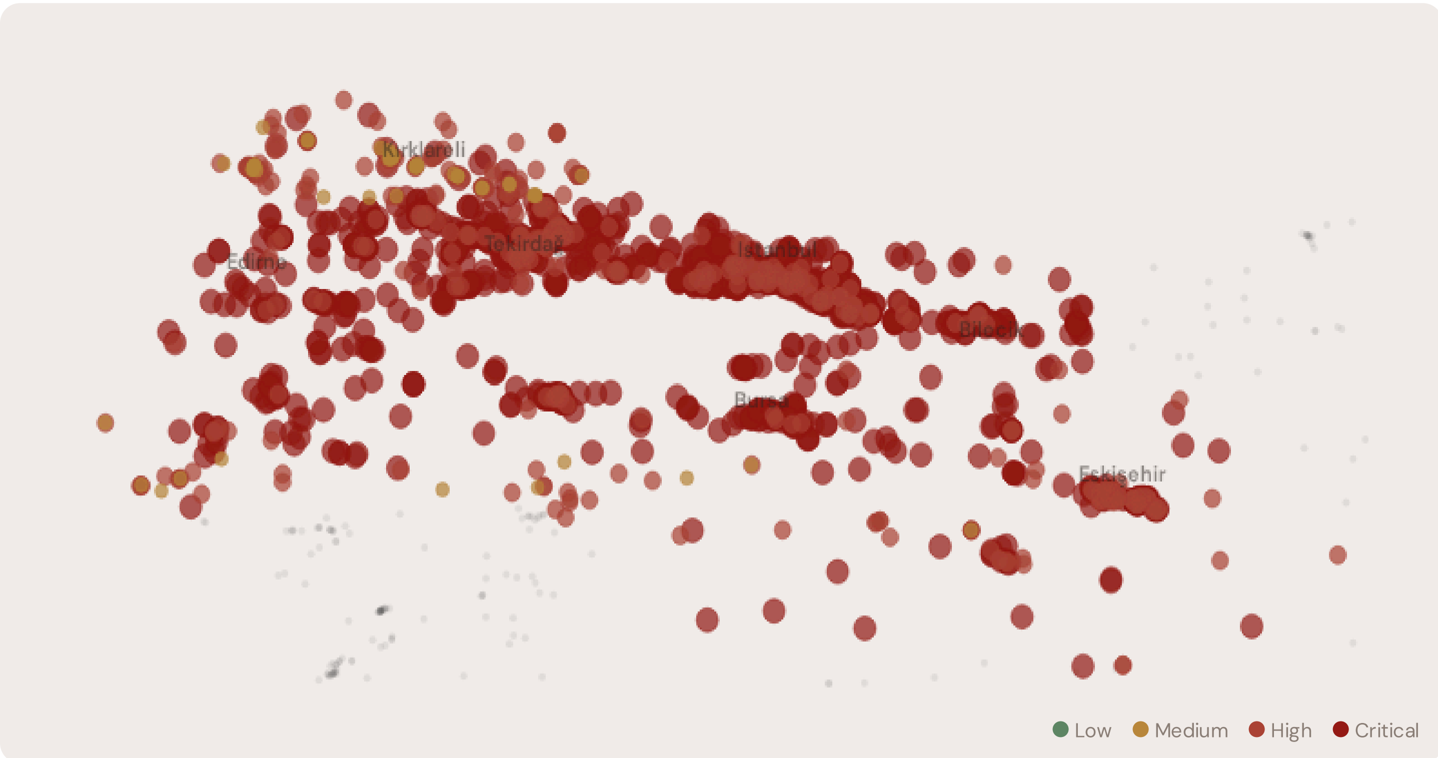
WORST İL

Tekirdağ

Avg R: 0.860 · 803 substations

Marmara hosts **2214 substations** across 7 provinces, making it the largest regional fleet in the SSI dataset. However, its risk profile is disproportionately concentrated in the upper bands: **578 substations (26.1%)** are classified High and **1584** Critical, with only **0** in the Low band. 75% of Marmara substations score above the national fleet median of 0.732. The corridor median R of 0.839 places Marmara among the most stressed regions in Turkey, driven by the combination of high continuity-of-supply deficits, ageing infrastructure, and accelerating DER penetration under TEDAŞ modernisation plans.

D.2 Corridor Map and Scoring



SUBSTATION	İL	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Substation TR-TR-0027	Istanbul	1.000	Critical	0.660	0.900	1.000	0.560	0.760	0.371	V, I
Substation TR-TR-0055	Bursa	1.000	Critical	0.660	0.900	0.952	0.560	0.760	0.431	V, I
DM	Tekirdağ	1.000	Critical	0.680	0.900	0.867	0.560	0.760	0.377	V, I
Substation TR-TR-0171	Edirne	1.000	Critical	0.710	0.900	1.000	0.560	0.760	0.302	V, I

SUBSTATION	İL	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Substation TR-TR-0226	Bursa	1.000	Critical	0.670	0.900	1.000	0.560	0.680	0.309	V, I
Substation TR-TR-0227	Bursa	1.000	Critical	0.660	0.900	0.885	0.560	0.760	0.453	V, I
Bursa Snayi 380kV TM	Bursa	1.000	Critical	0.670	0.900	1.000	0.560	0.760	0.362	V, I
Substation TR-TR-0250	Istanbul	1.000	Critical	0.680	0.900	1.000	0.560	0.760	0.312	V, I
DM22160	Istanbul	1.000	Critical	0.670	0.900	1.000	0.560	0.680	0.424	V, I
Bursa Sanayi 154kV TM	Bursa	1.000	Critical	0.660	0.900	0.978	0.560	0.760	0.335	V, I

... 2204 additional substations — explore full corridor in Digital Twin →

D.3 Component Analysis

COMPONENT AVERAGES — MARMARA VS FLEET

C — Continuity 0.553 vs 0.524 (+6%)

I — Infrastructure 0.724 vs 0.719 (+1%)

S — Saturation 0.626 vs 0.510 (+23%)

V — Voltage 0.572 vs 0.611 (-6%)

E — Economic 0.560 vs 0.403 (+39%)

T — Transition 0.351 vs 0.269 (+31%)

MODIFIER AVERAGES — MARMARA VS FLEET

R3 Consequence	1.134	fleet 1.101	↑
R4 Graph Crit.	1.008	fleet 1.008	→
R6a Restoration	1.040	fleet 1.031	↑
R7 Digital Read.	1.008	fleet 1.006	→

The dominant risk driver across the Marmara corridor is **T (Transition)**, averaging 0.351 against a fleet average of 0.269 — occupying 703% of its weight ceiling (w = 0.05). The second contributor is **V (Voltage)** at 0.572 vs fleet 0.611 (572% of ceiling). Among modifiers, the R3 consequence multiplier averages 1.134 (fleet: 1.101), reflecting the socio-economic fragility of Marmara's service-oriented, tourism-dependent economy. The R6b seismic modifier averages 1.131, capturing the region's moderate but non-negligible seismic exposure — a dimension invisible to every competing framework.

D.4 Province Breakdown

İL RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.

Tekirdağ 803 subs · avg R = 0.860 · H:106 C:692 M:5

Istanbul 525 subs · avg R = 0.860 · H:82 C:443 M:0

Bilecik	114 subs · avg R = 0.856 · H:18 C:95 M:1
Bursa	134 subs · avg R = 0.845 · H:22 C:109 M:3
Edirne	178 subs · avg R = 0.769 · H:75 C:91 M:12
Kırklareli	204 subs · avg R = 0.696 · H:99 C:74 M:31
Eskişehir	256 subs · avg R = 0.689 · H:176 C:80 M:0

The province-level breakdown reveals significant intra-regional variation. **Tekirdağ** leads with a mean R of 0.860 across 803 substations, while **Eskişehir** scores 0.689 — a spread of 0.171 within a single region. This disparity underscores why regional-level metrics (as used by CEER and JRC) mask the true distribution of grid stress. The SSI's substation-level granularity reveals that Marmara is not uniformly stressed but contains distinct corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications

1944 Marmara substations score R ≥ 0.650, placing them in the upper quartile of national risk. For grid investment prioritisation, this analysis identifies the Tekirdağ corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve — characterised by service-economy dependence, tourism exposure, and above-average energy poverty — are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

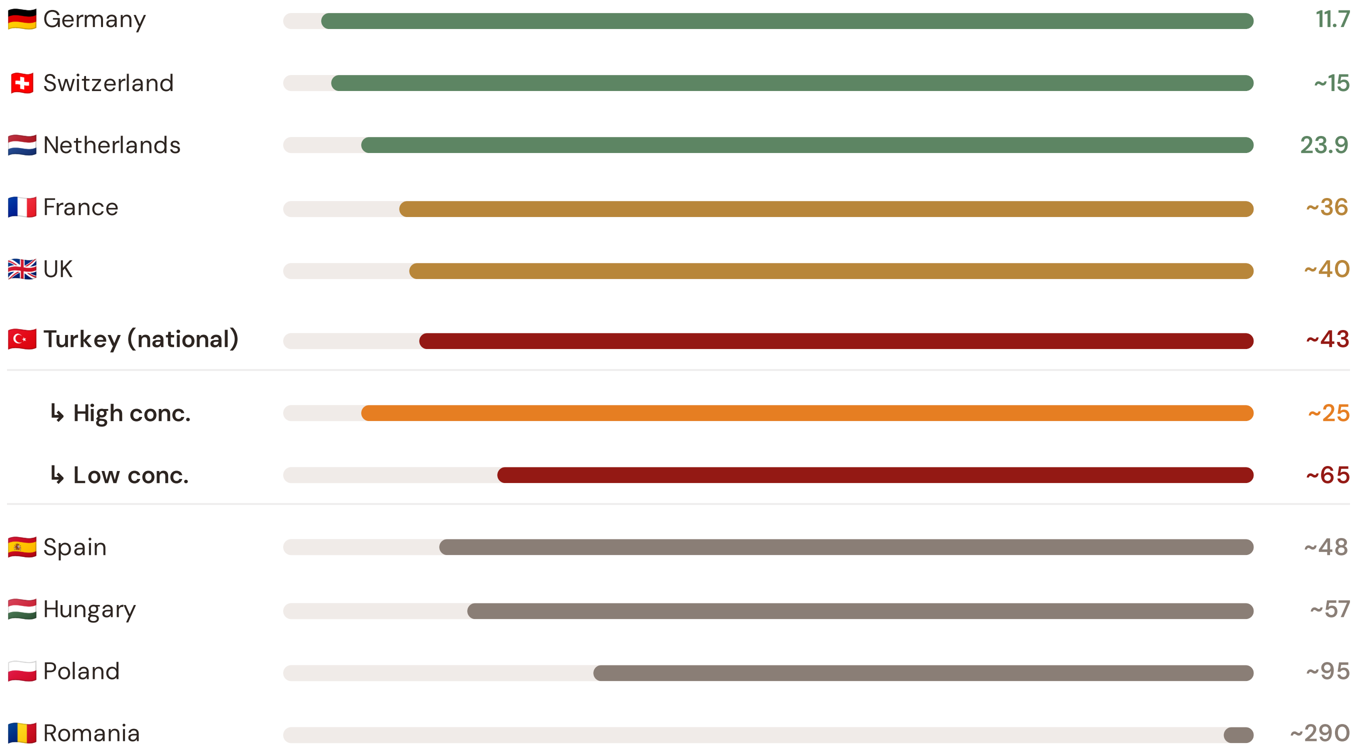
SECTION E — RECURRING MODULE

European Context Card

> **How this works:** Each edition includes one European Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the Marmara corridor analysis hinges on Turkey's continuity performance relative to European peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Germany (003), CMIP6 climate hazard vs Nordic peers (004), etc. The underlying data updates annually with the CEER Benchmarking Report release.

Unplanned SAIDI (min/customer/year) — Selected Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



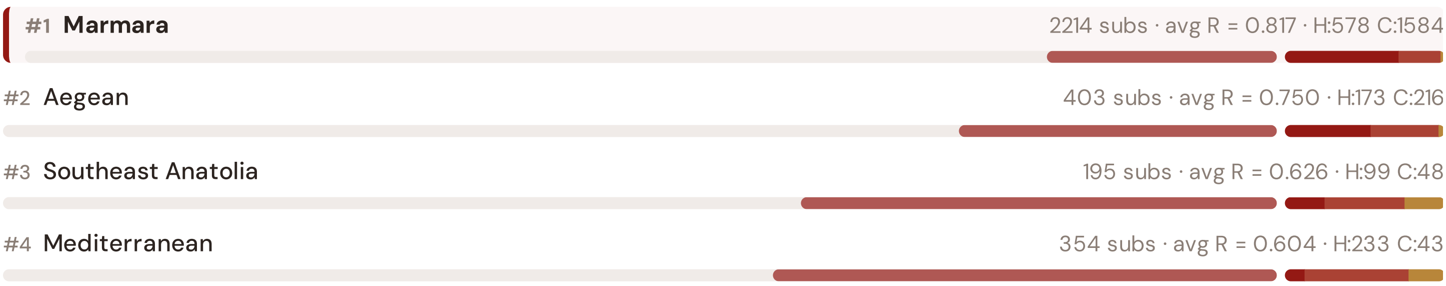
Source: CEER 7th Benchmarking Report (2022); Bundesnetzagentur (2024); EPDK (2023). Turkey sub-categories from EPDK territory classification. · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually with CEER BR release.

This month's key insight: The Marmara corridor deep-dive shows **2214 substations** (of 2214) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with EPDK's lower-reliability territory class. Marmara's average C of 0.553 is **1.1× the fleet average** of 0.524. In European terms, these substations individually perform closer to Hungarian or Spanish SAIDI levels (~48–57 min/customer/year) than to Turkey's own national average (~43 min). The SSI reveals what aggregate CEER statistics conceal: that Turkey's mid-tier European position is not a national condition but a statistical average of Dutch-quality urban performance and Eastern-European-level rural performance — and the Marmara corridor concentrates the worst of the latter.

Bölgeal Risk Ranking — All 20 Turkish Bölges

Where does Marmara sit within the national landscape? Bölges sorted by mean R_median, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.



#5 Central Anatolia 513 subs · avg R = 0.552 · H:327 C:17

#6 Black Sea 193 subs · avg R = 0.551 · H:95 C:21

#7 Eastern Anatolia 220 subs · avg R = 0.424 · H:40 C:1

Marmara ranks **#1 of 7 regions** by mean R_median, placing it among the upper tier of national risk. The three most stressed regions are Marmara, Aegean, Southeast Anatolia, while the three most resilient are Eastern Anatolia, Black Sea, Central Anatolia. 2 of 7 regions score above the fleet average of 0.716. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress across Turkey. It is precisely this substation-level granularity that positions the SSI as a tool for grid investment prioritisation: not treating "Eastern Turkey" as a monolith, but identifying the specific corridors within each region that demand attention.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the DNO and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.

Layer 1 — Data	Layer 2 — Valuation	**Layer 3 — Neural Net**	Layer 4 — Coverage	Layer 5 — Platform
SSI v4.0 component taxonomy, normalisation, MC engine	DNO + Community value stacks, TCO, real options	GBT+DNN ensemble, transfer learning, SHAP	Country prioritisation, DCI, registry, calibration	Products, API, revenue model, unit economics

LAYER 3 §0.5.5 · edition · May 2026

Uncertainty Quantification

Investment-grade predictions require calibrated confidence intervals, not just point estimates

A point estimate without uncertainty bounds is unusable for investment-grade analysis. The SSI-ENN produces calibrated P10/P50/P90 prediction intervals using two complementary approaches: quantile regression (the DNN head directly outputs multiple quantiles, trained with pinball loss) and conformal prediction (a distribution-free method that provides guaranteed coverage). The SSI's own Monte Carlo confidence intervals (CI_width, P5/P95) serve as the ground truth for calibration: the neural network's uncertainty estimates are validated against the analytical uncertainty from the 10,000-iteration copula engine. If the NN says "P90 NPV = ₺4.2M" for a substation, that bound must be calibrated — meaning 90% of true valuations (from the full analytical model) should indeed fall below ₺4.2M. This calibration is verified per country, per voltage class, and per risk band.

KEY CONCEPTS

- → Quantile regression (pinball loss) + conformal prediction
- → Calibrated against SSI Monte Carlo ground truth
- → Per-country, per-voltage, per-band calibration verification
- → Investment-grade: P10/P50/P90 for all outputs

LIVE SSI DATA CONNECTION

The Turkish training set comprises 4,092 substations with complete 95-feature vectors. Average CI_width of 0.183 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 2 Low, 615 Medium, 1545 High, 1930 Critical — ensures balanced training across all risk bands.

Coming next month: LAYER 3 Explainability & SHAP Values — *Clients need to understand not just the prediction but WHY a substation ranks highly*

SECTION G

Looking Ahead

NEXT MONTH — EDITION 003

The Mersin–Adana Industrial Belt

Deep-dive into Akdeniz's grid risk profile — substation map, component analysis, province breakdown. European Context Card: Petrochemical + agricultural load concentration and EV transition readiness.

NEXT DATA REFRESH

Q3 2026 — 5 Quarterly Sources

EDAS-OSM, EGOV-REG, AEDAS-DGSI, TUBITAK-R, ETKB-REN are due for refresh in Q3. Impact: DER registry updates (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **EPDK (Enerji Piyasası Düzenleme Kurumu)** (8 variables → C1–C4 (SAIDI/SAIFI), I1–I3 (IRI), quality regulation).

FLEET SPOTLIGHT

0 Substations Approaching Critical Degradation

The SSI's 4-state Markov degradation model identifies **0 substations** with a stationary critical probability above 70% — meaning their long-term equilibrium state is dominated by degraded or critical condition. Without intervention, these substations will trend toward failure. Average estimated time-to-critical: ? years.

Edition 003 will be published on the second Thursday of July 2026. Deep-dive region: **Akdeniz**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent an Turkish utility, EPDK, municipal energy company, or academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Edition 002

Published 14 May 2026 · SSI Index v4.0.2 · 4,092 substations · 14,221 grid lines · 6 components · 20 metrics · 6 modifiers

10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5